

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. Computer Engineering Sem - I University Examination, 2010-11

CE 704 Soft Computing

Date: 12.05.11, Thursday Time: 10:00 a.m To 01:00 p.m Maximum Marks: 70

- Instructions:** (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main supplementary, don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions.

1. What is neural network? What are the benefits of Neural Network? What are various network architectures? 3
2. What is soft computing? How the hybridization of various technologies can be done? What are the applications of soft computing? 3
3. Why a single layer of perceptron cannot be used to solve linearly inseparable problems? 3
4. Define Delta rule and give its uses in Back Propagation algorithm. 2

Q-2

- [A] Discuss the BP NN algorithm along with its mathematical derivation. 4

OR

- [A] Explain the architecture and training of Kohonen's self-organizing network. 4
[B] How Fuzzy Logic is different from conventional control System? 4
[C] "Increasing the number of hidden layer neurons gives a better function approximation". Is this statement correct? Justify your answer. 4

OR

- [C] Describe the working of biological Neuron. Compare it with artificial Neuron. 4
Describe the advantages of Neural Network with conventional techniques.

Q-3

- [A] Discuss briefly about the limitations of back propagations. 4

OR

- [A] Implement a back propagation algorithm to solve the logic of AND Gate problem. 4
[B] Discuss the advantages of fuzzy logic over Crisp Logic. Describe at least three characteristics to classify birds. Define fuzzy set for each characteristic and give example of birds that can be classified using these sets. 4
[C] Explain the steps in the solution of a general optimization problem, by a NN and a Genetic Algorithm. 4

OR

- [C] Why mutation and fitness function is required? How selection of candidate mating pool will affect convergence of algorithm? 4

SECTION-II

Q-4

1. Let Y and R be two fuzzy sets of young and rich people. What is the member grade of a person being young and rich if the member grade of a person being young is 0.8 & being rich is 0.7? 3
2. Which are the heuristic for making back propagation better? 4
3. Explain the steps in the solution of a general optimization problem. 4

Q-5

- [A] Implement a back propagation algorithm to solve EX-OR problem and try the architecture in which there is a hidden layer with three hidden units where the network is fully connected. 4
- [B] Define Genetic Algorithm. What are the drawbacks of GA. List the Application of GA. 4
- [C] What are the stopping criteria while executing GA? How can we improve the Performance of Genetic Algorithm? 4

OR

- [A] How GA play important role with integration of other technology? Explain in details. 4
- [B] What do you mean by Fuzzy decision making? How it differs from Fuzzy Bayesian decision making? 4
- [C] Give a model for a neuron of Radial Basis function network. 4

Q-6

- [A] How is Boltzman learning rule applied to Energy minimization? 4
- [B] Describe the architecture and operation of ART network. 4

OR

- [B] Describe various crossover techniques for GA. How crossover rate is defined? 4
- [C] Explain Fuzzy Inference System in detail with its block diagram. 4

OR

- [C] Define Encoding System in GA. Write the Encoding representation of Transportation Problem. 4